

Curl and Divergence of Vector Field:

Chapter 09
9.7, 9.8, 9.9

Gradient (Gradient of scalar ϕ is vector field)

Gradient of scalar function is

$$\vec{\nabla} \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \quad (1)$$

where $\phi = \phi(x, y, z) \quad (2)$

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \quad (3)$$

Ex Find gradient of following function.

(1) $\phi(x, y) = x^2 \sin(5y) \quad (4)$

(2) $\phi(x, y) = ze^{-xy} \quad (5)$

Sol (1) $\phi(x, y) = x^2 \sin(5y)$

$$\begin{aligned} \vec{\nabla} \phi &= \frac{\partial}{\partial x} (x^2 \sin(5y)) \hat{i} + \frac{\partial}{\partial y} (x^2 \sin(5y)) \hat{j} \\ &= \left((2x) \sin(5y) \right) \hat{i} + \left(5x^2 \cos(5y) \right) \hat{j} \end{aligned} \quad (6)$$

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Q2 $\phi(x, y, z) = z e^{-xy}$ — (7)

$$\nabla \phi = \frac{\partial}{\partial x} (z e^{-xy}) \hat{i} + \frac{\partial}{\partial y} (z e^{-xy}) \hat{j} + \frac{\partial}{\partial z} (z e^{-xy}) \hat{k}$$

$$\nabla \phi = (z e^{-xy} (-1)) \hat{i} + (z e^{-xy} (-x)) \hat{j} + (e^{-xy}) \hat{k}$$

— (8)

Divergence of vector field:-

is denoted and defined as

$$\text{div } \vec{F} = \vec{\nabla} \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (P \hat{i} + Q \hat{j} + R \hat{k})$$

$$\boxed{\text{div } \vec{F} = \vec{\nabla} \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}} \quad \text{--- (1)}$$

Curl of vector field:-

is denoted and defined as

$$\text{Curl } \vec{F} = \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \quad \text{--- (2)}$$

Example #01

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Compute $\text{div } \vec{F}$ and $\nabla \times \vec{F}$ of

$$\vec{F} = (x^2y)\hat{i} + (xyz)\hat{j} - (x^2y^2)\hat{k} \quad \text{--- (1)}$$

Sol

$$\vec{F} = P(x,y,z)\hat{i} + Q(x,y,z)\hat{j} + R(x,y,z)\hat{k} \quad \text{--- (2)}$$

01

$$P(x,y,z) = x^2y, \quad Q = xyz, \quad R = -x^2y^2$$

(3) (4) (5)

02

$$\text{div } \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \quad \text{--- (6)}$$

$$= \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(xyz) + \frac{\partial}{\partial z}(-x^2y^2)$$

(7)

$\text{div } \vec{F} = 2xy + xz + 0$

 (8)

Q3

⑥

$$\text{Curl } \vec{F} = \nabla \times \vec{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & xyz & -x^2y^2 \end{vmatrix}$$

$$= \hat{i} \left(\frac{\partial}{\partial y} (-x^2y^2) - \frac{\partial}{\partial z} (xyz) \right)$$

$$- \hat{j} \left(\frac{\partial}{\partial x} (-x^2y^2) - \frac{\partial}{\partial z} (x^2y) \right)$$

$$+ \hat{k} \left(\frac{\partial}{\partial x} (xyz) - \frac{\partial}{\partial y} (x^2y) \right)$$

$$= \hat{i} (-2x^2y - xy)$$

$$- \hat{j} (-2xy^2 - 0)$$

$$+ \hat{k} (yz - x^2)$$

Ex Find $\text{div } \vec{F}$ and curl $\vec{F}$ of

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$$\vec{F} = yz^2 \hat{i} + xy \hat{j} + yz \hat{k} \quad \text{--- (1)}$$

Sol

$$\text{Sol} \quad P = yz^2, \quad Q = xy, \quad R = yz$$

$L(2) \quad L(3) \quad L(4)$

$$\text{Sol} \quad \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz^2 & xy & yz \end{vmatrix} \quad \text{--- (5)}$$

$$= \hat{i} \left(\frac{\partial}{\partial y} (yz) - \frac{\partial}{\partial z} (xy) \right) - \hat{j} \left(\frac{\partial}{\partial x} (yz) - \frac{\partial}{\partial z} (yz^2) \right) + \hat{k} \left(\frac{\partial}{\partial x} (xy) - \frac{\partial}{\partial y} (yz^2) \right) \quad \text{--- (6)}$$

$$= \hat{i} (z) + \hat{j} (2z) + \hat{k} (y - z^2) \quad \text{--- (7)}$$

$$\text{Sol} \quad \text{div } \vec{F} = \frac{\partial}{\partial x} (yz^2) + \frac{\partial}{\partial y} (xy) + \frac{\partial}{\partial z} (yz) \quad \text{--- (8)}$$

$$= 0 + x + y = x + y \quad \text{--- (9)}$$

Application

- (1) if $\nabla \times \vec{F} = 0$, then the given vector field is conservative
- (2) if $\nabla \times \vec{F} = 0$, then fluid is called as irrotational.
- (3) if $\text{div } \vec{F} = 0$, then the given fluid is incompressible

Exercise

Q4 - Q13 (9.9) $\rightarrow$ (curl)

Q1 - Q4 (9.8) $\rightarrow$ (Divergence)

Q1 - 6 (9.7) $\rightarrow$ (Gradient)
Q11 - Q16